

Category	Item	Description	Quantity	Unit	Purpose
Core	USR-M100-ETH	Industrial Remote I/O Edge Gateway (Ethernet Version)	1	pcs	Replaces ESP32; handles DI and PZEM
	Mean Well MDR-20-24	DIN Rail Power Supply (Input: 100-240VAC Output: 24V DC / 1A)	1	pcs	Powers M100 and sensors
	TTL to RS485 Isolated Converter	Industrial-grade Serial converter (XY-017)	1	pcs	Required for PZEM-004T (TTL version)
	PZEM-016 (Alternative)	Modbus-RTU Power Meter (Native RS485)	1	pcs	Recommended alternative to PZEM-004T
Cable	Cat6 Ethernet Cable	RJ45 Patch Cord (shielded)	1	pcs	Wired connection to Server
	RS485 Shielded Cable	2-core twisted pair (RVVP 2x0.5mm)	2	meters	Connection between PZEM and M100
	Signal Wire	0.5mm2 Multi-strand (Red/Black/Blue)	5	meters	Machine signals (GOOD/NG) to DI
Mechanical	35mm DIN Rail	Aluminum/Steel mounting rail	20	cm	Mounting for M100 and PSU
	Bootlace Ferrules	Wire terminal crimps (0.5mm2 - 1.5mm2)	1	set	Secure terminal connections
	Industrial Enclosure	IP65 Box / Panel	1	pcs	Protection from factory environment